

# Open Source EDA Tools In Fedora Electronic Lab

Design circuits easily with software applications for schematic capture, simulation, layout, numerical computations and more!

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**T**oday, a variety of electronic design automation (EDA) tools are available commercially. This month's DVD contains Open Source versions of EDA tools, which Fedora has included in the Fedora Electronic Lab. These tools provide the same functionality as their priced counterparts and can be used in laboratories for research as well as in companies for production.

## Alliance

Alliance is a complete set of free computer-aided design (CAD) tools and portable libraries for very-large-scale integration (VLSI) design. It includes a VHDL compiler and simulator, logic synthesis tools, and automatic place-and-route tools. A complete

set of portable CMOS libraries is provided including a RAM generator, a ROM generator and a data-path compiler.

Alliance provides CAD tools covering digital design flow functions including VHDL compilation and simulation, model checking and formal proof, register-transfer level and logic synthesis, data-path compilation, macro-cells generation, place and route, layout edition, netlist extraction and verification.

Microsoft version downloadable at: [www.layouteditor.net](http://www.layouteditor.net)

Homepage: [www.layouteditor.net/filedownload.php5?fileid=1224](http://www.layouteditor.net/filedownload.php5?fileid=1224)

## Electric VLSI Design System

It is an electrical CAD system that can handle many forms of circuit design including custom ASIC layout, schematic capture (digital and analogue), hardware description language specifications and electro-mechanical hybrid layout.

The CAD operations of Electric include design rule check, electrical rule check, simulation and simulation interface, rout-

ing, VHDL compilation, silicon compilation, network consistency check, logical effort analysis and project management.

The file formats supported by Electric include CIF, GDS, EDIF, DXF, SUE, VHDL, Verilog, EAGLE, PADS, ECAD, PostScript, HPGL and PNG.

Electric is compatible with Unix, Windows and Mac platforms.

Microsoft version downloadable at: <http://ftp.gnu.org/pub/gnu/electric/electric-7.00.tar.gz>

Homepage: [www.gnu.org/s/electric](http://www.gnu.org/s/electric)

## gEDA

gEDA project offers a mature suite of free software applications for electronics design including schematic capture, attributed management, bill-of-materials generation, netlisting into over twenty netlist formats, analogue and digital simulation, and PCB layout. It is compatible with Unix, Windows, Mac and Linux platforms.

Within the gEDA Suite, gEDA/gaf ('gaf' stands for 'gschem and friends') is the smaller subset of tools grouped under the gEDA name and maintained directly by the gEDA project's founders.

GEDA/gaf includes:

**gnelist.** It is a netlist generation program which takes gEDA/gaf schematic file and the required symbol files as input and converts these into netlists.

**gysmcheck.** It is a syntax checker for schematic symbols. Give it a symbol file and it will go through and verify that the symbol will work in gaf.

**gattrib.** It is a spreadsheet program for editing symbol attributes on a sche-

